

# Structure-aware controls for topological data analysis of perturbation single-cell RNA-seq: a dispersion-matched null, a traversal test, and a cautionary case study

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<sup>1</sup>Preprint — methods report. Code: <https://github.com/htlin222/sctda-cancer-plasticity>

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## Abstract

As topological data analysis (TDA) enters single-cell biology, scalar summaries of persistent homology — most simply the maximum degree-1 persistence ( $\max\text{-}H_1$ ) — are beginning to be read as quantitative measures of “cyclic” cell-state structure and compared across conditions. This is hazardous for two reasons that are individually known but not jointly controlled in practice: Vietoris–Rips persistence magnitude grows with the spread (dispersion) and sampling of a point cloud even in pure noise [3], and the cell cycle is an intrinsic loop in expression space [1,5]. We assemble four controls into one reusable significance procedure for perturbation scRNA-seq — (i) a **covariance-matched Gaussian null** preserving transcriptional dispersion; (ii) a **circular-coordinate traversal test** that asks whether a detected loop is occupied before any cyclic reading; (iii) **cell-cycle regression evaluated relative to the dispersion null** rather than on raw persistence; and (iv) **bootstrap directional testing** — and validate it on synthetic data with known ground truth (sensitive to a genuine traversed loop, specific against benign dispersion and non-Gaussian non-cyclic structure). As a cautionary case study we report our own analysis of five longitudinal EGFR-mutant lung-cancer systems, in which an apparently compelling, reproducible, cross-scale drug-associated  $\max\text{-}H_1$  signal dissolves under the controls: cells never traverse the loop ( $R \geq 0.965$  vs 0.881 for a 12%-occupied control), baseline  $\max\text{-}H_1$  equals the dispersion null, and the in-vitro drug excess is the cell-cycle loop. The framework also isolates genuine structure where it exists — a drug-emergent multiciliated program in PDX residual disease — showing it passes real signal as well as rejecting confounded signal.

**Contributions and relation to prior work.** The individual confounds and several remedies are known; our contribution is their *integration into one corrected significance procedure for single-cell  $H_1$* , with a validated operating point, plus an honest worked failure.

1. A **covariance-matched Gaussian null** for  $\max\text{-}H_1$  on scRNA-seq. The dominant published scTDA null is within-cell gene-label shuffling [1], which destroys covariance and is beaten by essentially any real data; the principled TDA-theory alternative is subsampling confidence sets [4]. A dispersion-preserving null for single-cell persistence appears not to have been used before, though weak-vs-strong-null critiques exist for embedding geometry [9].
2. A **circular-coordinate traversal test** [10] establishing loop occupancy as a precondition for any “cyclic” interpretation.
3. Made explicit: a **cell-cycle control based on raw  $\max\text{-}H_1$  surviving S/G2M regression is inadequate**, because raw persistence is dispersion-dominated [3]; the cell-cycle contribution must be judged relative to the dispersion null. The cell-cycle-as-loop confound is established [1,5]; the inadequacy of the raw-persistence control, to our knowledge, is not.

4. A **synthetic benchmark** (sensitivity/specificity) and a five-system **cautionary case study** — our own — of a confounded signal that is reproducible, cross-scale, and wrong.

## 1 Introduction

Topological data analysis is entering single-cell biology, with reviews now cataloguing applications of persistent homology and Mapper to scRNA-seq [8]. As the field matures, a natural step is to summarise topology with a scalar —  $\max\text{-}H_1$ , total persistence, persistence entropy — and compare it across conditions, reading an increase as more cyclic structure or plasticity. We argue this step needs controls current practice does not jointly apply. Two confounds are individually documented. First, Vietoris–Rips persistence is sensitive to spread and sampling: the maximally persistent cycle of pure random points grows with sample size [3], raw magnitude is metric/scale-dependent (motivating stabilised summaries [6,7]) and outlier-sensitive (motivating density-robust filtrations [8b]). Second, the cell cycle traces a circle in expression space [5] and is the canonical  $H_1$  generator in scRNA-seq; the field-defining scTDA method states the cell cycle “will give rise to periodic structures in the expression space” [1]. The common nulls do not control the first (gene-label shuffling [1] destroys dispersion), and a recent review notes single-cell TDA “may not provide rigorous statistical assessments” [8].

This is a *preventive* contribution, not a claim of widespread error: scalar persistence-vs-condition comparison on scRNA-seq is still uncommon (established scTDA methods use Mapper graphs and per-cell features [1,2]), and careful applications controlling sample size with diagram distances exist [11]. Our point is that as the scalar-readout step becomes attractive, these controls should accompany it — and that the trap is real, because we fell into it ourselves. We present our own five-system analysis as the cautionary example.

## 2 Results

### The control framework

Per condition, report: (1) *dispersion test*, observed  $\max\text{-}H_1$  over the  $\max\text{-}H_1$  of a Gaussian matched to the embedding’s mean and covariance (ratio  $\sim 1$  = dispersion; robustly  $> 1$  = structure beyond dispersion); (2) *traversal test*, the Rayleigh  $R$  of circular coordinates [10] ( $R \sim 1$  = loop not occupied); (3) *cell-cycle test*, the ratio after S/G2M regression [12]. All bootstrapped [4]. Decision: structure if ratio robustly  $> 1$ ; traversed loop only if also  $R < 0.6$ ; cell-cycle driven if the excess does not survive regression.

### Sensitive and specific on synthetic ground truth (Fig. 3)

Gaussian blob: ratio 0.93,  $R = 0.99$  (no structure). Traversed loop: ratio 8.35,  $R = 0.42$  (traversed loop; sensitivity). Sparse arc: ratio 1.54,  $R = 0.96$  (structure, not traversed). Two clusters (non-Gaussian, non-cyclic): ratio 0.93 (no structure; specificity — the Gaussian null does not false-positive on benign non-Gaussianity). A per-PC permutation null agrees and is more conservative.

### Case study and Control 1 — cells do not traverse the loop

$\max\text{-}H_1$  rises with drug in osimertinib cell line and PDX (D0→D14 1.41 → 3.78; PDX 2.79 → 5.46; Fig. 1); erlotinib is already weak/non-monotonic (D9 1.77, D11 1.08). Circular coordinates collapse to one angle in every cohort:  $R \geq 0.965$  (Fig. 2a) — more concentrated than the  $R = 0.881$  12%-occupied control. Clonal lineages are *more* concentrated than random. The loop is sparse and unoccupied.

## Control 2 — baseline max- $H_1$ is dispersion

Against the Gaussian null, baseline max- $H_1$  does not exceed dispersion (osimertinib D0/D3 ratio  $\approx 0.9$ ; observed  $>$  null in 18–31% of subsamples; Fig. 1); an excess appears only under drug (D7/D14 in 100%/98%; PDX residual 99%). A Gaussian preserves only second moments, so its failure mode is conservative for a negative conclusion.

## Control 3 — in-vitro excess is the cell cycle; in-vivo excess is not

The osimertinib D14 excess is proliferating cells (DE: PTTG1, CKS1B, MYBL2; MKI67/PCNA/CCNB1/CDK1/BIRC5) and does not survive S/G2M regression (ratio 1.18 $\rightarrow$ 0.99; D14 $>$ D0 in only 31% of bootstraps; Fig. 2b). The common control — raw max- $H_1$  preserved after regression — would have wrongly concluded “not cell cycle,” because raw max- $H_1$  is dispersion-dominated [3]. The framework decomposes: in the in-vivo PDX the residual excess is *not* cell cycle (G1-only ratio 2.07, 90% CI [1.37, 2.61],  $> 1$  in 100% of bootstraps), yet still fails the traversal test ( $R = 0.965$ ). Across all five systems: dispersion = baseline; cell cycle = in-vitro excess; in-vivo residual = genuine but non-traversed.

## A confound-resistant clonal statistic is under-powered

$M(t) = 1 - H_{\text{obs}}/H_{\text{null}}$  against a lineage-shuffle null is non-robust:  $M(\text{D0})$  point estimate 0.48 but 90% CI  $[-0.05, 0.29]$ ; decay sign flips across settings (positive in 3/9), non-monotonic, halved by a depth-stratified null (Fig. 4). Cause: only 12–33 clones with  $\geq 3$  cells/timepoint.

## Beyond rejection: the framework isolates a genuine program

The PDX residual excess (surviving every control) is a coherent multiciliated airway program (PIFO, RSPH1, CFAP45/126/157, CAPS/CAPSL, CETN2, TPPP3; AGR2/AGR3), drug-emergent in YU-006 (0.1% $\rightarrow$ 9.6%;  $\approx 6 \rightarrow \approx 187$  cells while the tumour contracts threefold), consistent with the differentiated states EGFR-TKI persists adopt. Two bounds: model-specific (absent in YU-003); CNV could not resolve transdifferentiation vs an expanded non-malignant population without a matched-normal reference. The framework isolates real structure as well as rejecting confounded structure.

## 3 Methods

GEO GSE134839/150949/243562/131907 + Maynard 2020; scanpy QC/HVG/PCA (seed 42), subsampled 1,000–1,500 cells; **ripser** 0.6 Vietoris–Rips to  $H_1$  on top-30 PCs ( $\mathbb{F}_2$ ). Dispersion null: draws from  $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ , ratio = observed/median(null), 30 bootstrap subsamples + 90% CIs, second null by per-PC permutation. Traversal: **dreimac CircularCoords** [10], Rayleigh  $R$  (validated: noisy circle  $R = 0.12$ ; 12%-occupied loop  $R = 0.88$ ). Cell-cycle: leave-one-cluster-out, Wilcoxon DE, S/G2M regression [12] with the dispersion ratio bootstrapped. In-vivo identity: DE on raw counts, ciliated signature scored across PDX, CNV by **infercnvpy**. Code: **scripts/16\_\*.py**–**scripts/27\_\*.py** (MIT).

## 4 Discussion

A cross-system max- $H_1$  signal of drug tolerance is largely explained by dispersion and, in vitro, the cell cycle; genuine residual structure still does not evidence cyclic plasticity, because it is not traversed. That the signal looked compelling across five systems, survived a published-style cell-cycle control, and “replicated” in patient tumours is why these controls matter: a confounded statistic can be reproducible, cross-scale, and wrong. Our contribution is preventive and integrative. Each confound has prior literature — density/sample-size dependence [3], cell-cycle loops [1,5],

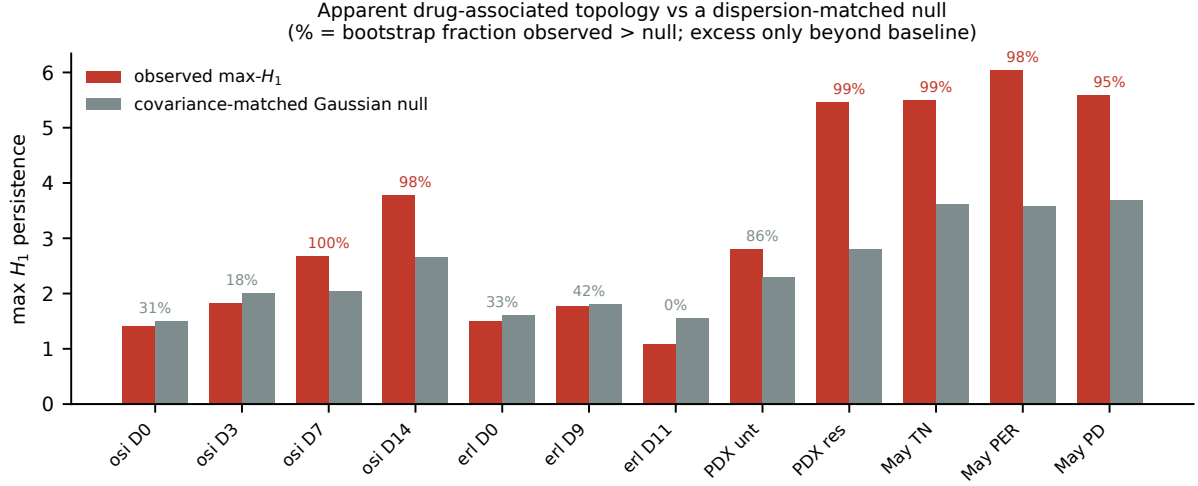


Figure 1: Observed max- $H_1$  vs a covariance-matched Gaussian (dispersion) null across five systems. Percentages: bootstrap fraction observed > null. Excess appears only under drug.

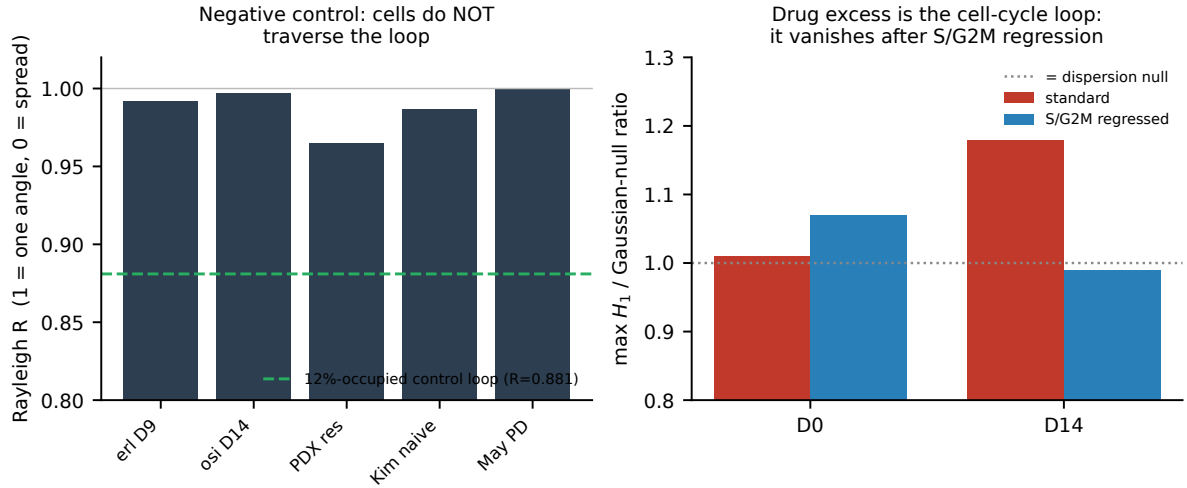


Figure 2: (a) Traversal test:  $R \geq 0.965$  in all cohorts vs 0.881 for a 12%-occupied control. (b) The in-vitro drug excess vanishes after S/G2M regression.

preprocessing sensitivity and missing significance tests in single-cell TDA [8], and, closest to our thesis, the demonstration that geometric structure significant under weak nulls vanishes under stronger nulls in single-cell foundation-model embeddings [9]. We differ in target and null (raw-expression  $H_1$  + dispersion-matched Gaussian null vs embedding geometry + rewiring nulls) and contribute the *integration* into one validated procedure. The framework also surfaces a drug-emergent multiciliated program in residual disease; whether it is tumour transdifferentiation conferring tolerance is a prospective, experiment-bearing programme.

**Outlook.** A discovery claim would require resolving cell-of-origin (matched-normal CNV or EGFR genotyping), reproduction in further models or patient residual disease, and functional testing of tolerance.

**Limitations.** Cell-cycle attribution is shown on the cell line; the PDX cell-of-origin is unresolved. We address max- $H_1$ ; the dispersion null should extend to other persistence summaries [6,7] and higher homology, untested here.

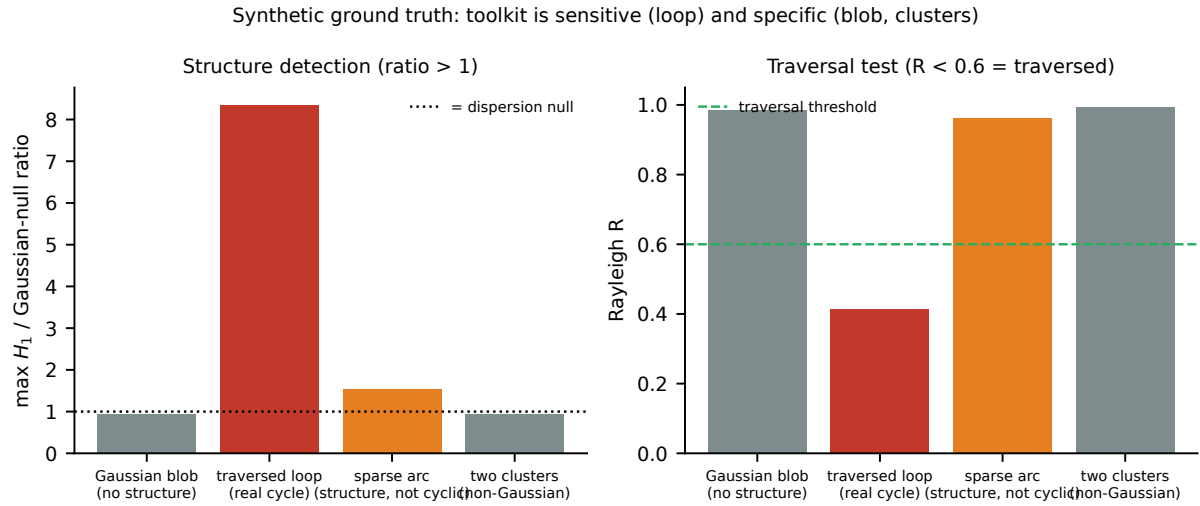


Figure 3: Synthetic ground truth: sensitive (passes the traversed loop) and specific (rejects the Gaussian blob and the two-cluster non-Gaussian case).

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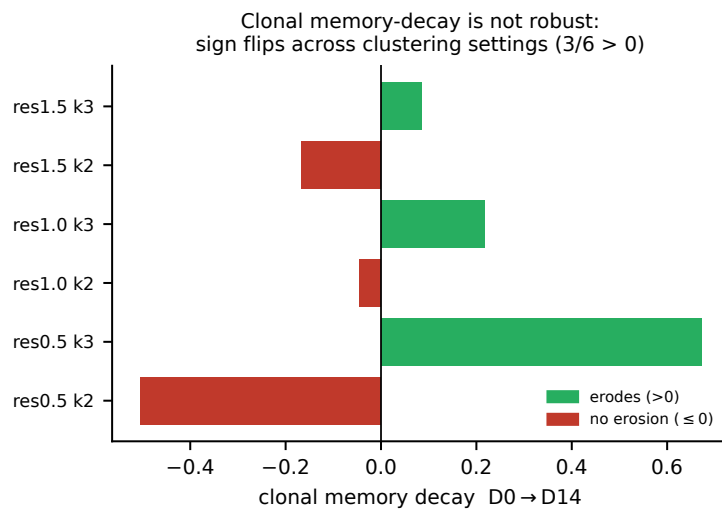


Figure 4: Clonal memory-decay is not robust: the sign of the D0→D14 change flips across clustering settings.